



Applied AI Mastery Program

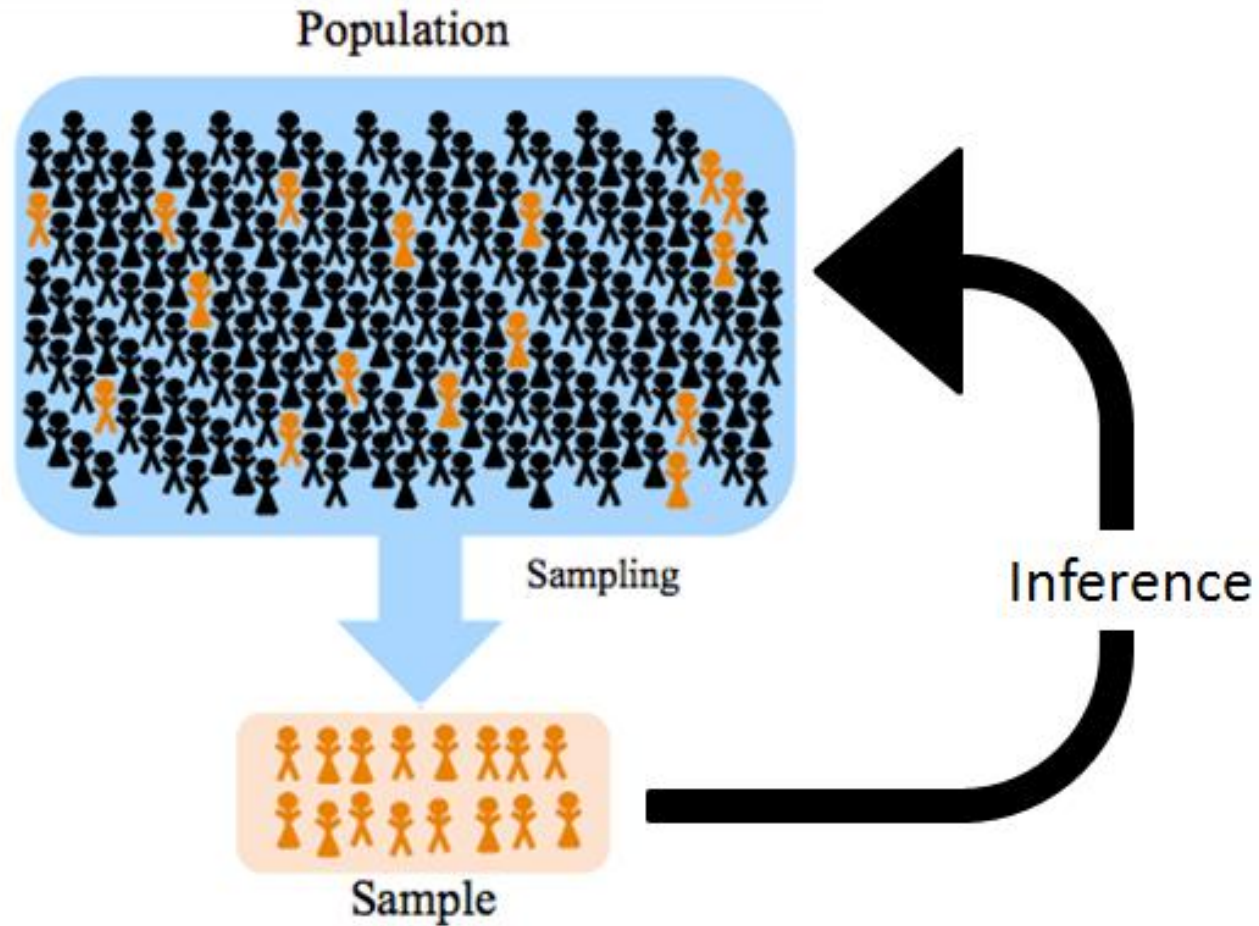
Module 1 – The Basics from the Minimalist Perspective

Rina Buoy, PhD



Minimum Statistics

Sample vs. Population



Source: <https://statsandr.com/blog/what-is-the-difference-between-population-and-sample/>

Are the custom data, used to a credit score model, sample or population?

Descriptive Statistics

- Mean
 - Median
 - Mode
- 
- Measures of Central Tendency
- Variance
 - Standard deviation
- 
- Measures of Variability
- Percentiles
- 
- Measures of Position

Mean (Arithmetic Average)

- **Definition:** The sum of all values in a dataset divided by the total number of values. It is highly influenced by outliers.
- **Population Formula:**

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

- **Sample Formula:**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Median

- **Definition:** The middle value of a dataset when the data is ordered from least to greatest. It is a robust measure that is not easily skewed by outliers.
- **Formula:** First, sort the dataset in ascending order, where n is the total number of observations.
- If n is **odd**, the median is the value at position: $\frac{n+1}{2}$
- If n is **even**, the median is the average of the two middle values at positions: $\frac{n}{2}$ and $\frac{n}{2} + 1$

Median

1, 3, 3, **6**, 7, 8, 9

Median = **6**

1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$
= **4.5**

Mode

- **Definition:** The value (or values) that appears most frequently in a dataset. A dataset can have one mode (unimodal), multiple modes (multimodal), or no mode at all if every value appears only once.
- **Formula:** There is no mathematical equation; it is simply the most frequent x_i in the set.

Variance

- **Definition:** A measure of how much the data points in a set are spread out from the mean. It represents the average of the squared differences from the mean.

- Population Formula:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

- Sample Formula:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard Deviation

- **Definition:** The average amount of variability in your dataset. It tells you, on average, how far each value lies from the mean. It is simply the square root of the variance, which brings the unit of measurement back to the original unit of the data.

- Population Formula:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

- Sample Formula:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Percentiles

- **Definition:** A value below which a specific percentage of the data falls. For example, if you score in the 90th percentile on a test, it means you scored higher than 90% of the people who took the test.

- **Formula:**

First, sort the data in ascending order. To find the k -th percentile, calculate the index (i):

$$i = \frac{k}{100} \times n$$

Percentiles

Data Set:

2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 8, 9, 9, 10, 11, 11, 12

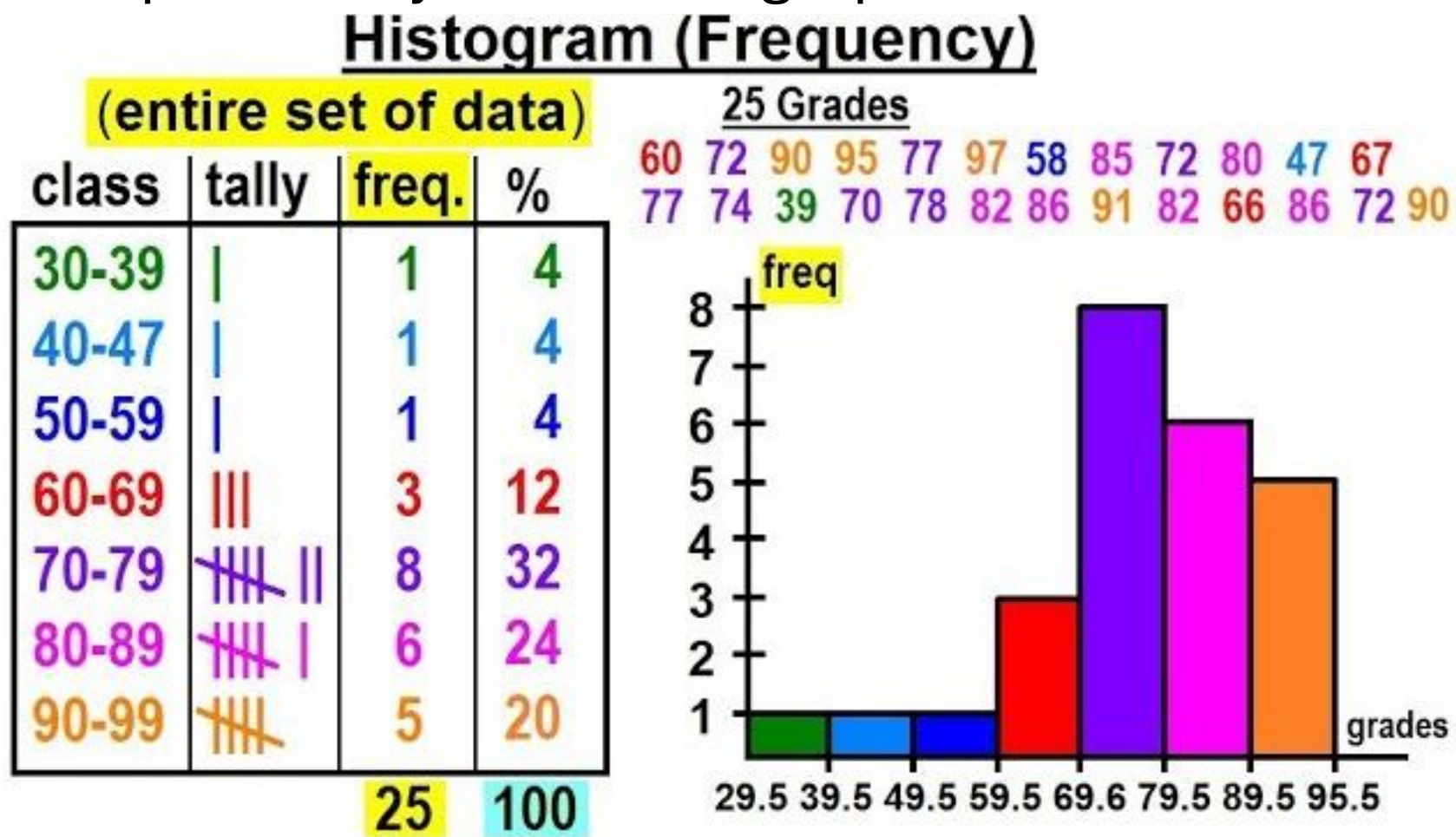
What is the **percentile** ranking of '10'?

$$\text{Percentile Rank of } x = \frac{\# \text{ of values below } x}{n} * 100$$

$$\text{Percentile Rank of '10'} = \frac{16}{20} * 100 = 80\%$$

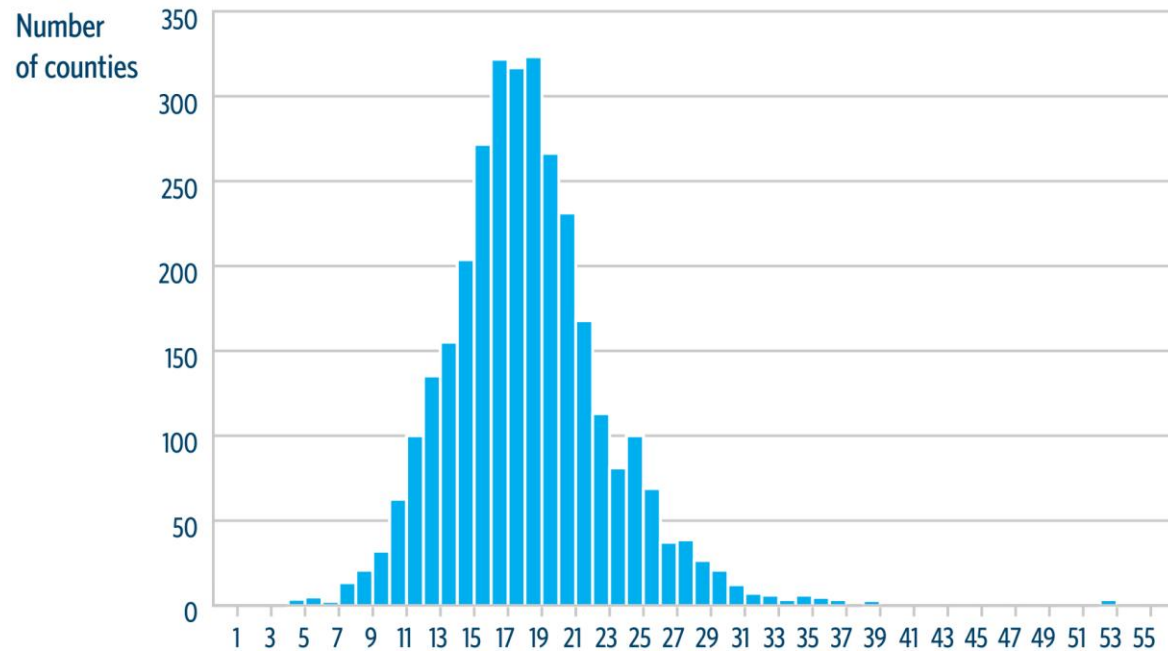
Data Distributions

- **Definition:** A listing, function, or graph that shows all the possible values (or intervals) of a dataset and how often they occur. It tells you the shape of your data and the probability of observing a particular value.



Modelling Data Distributions

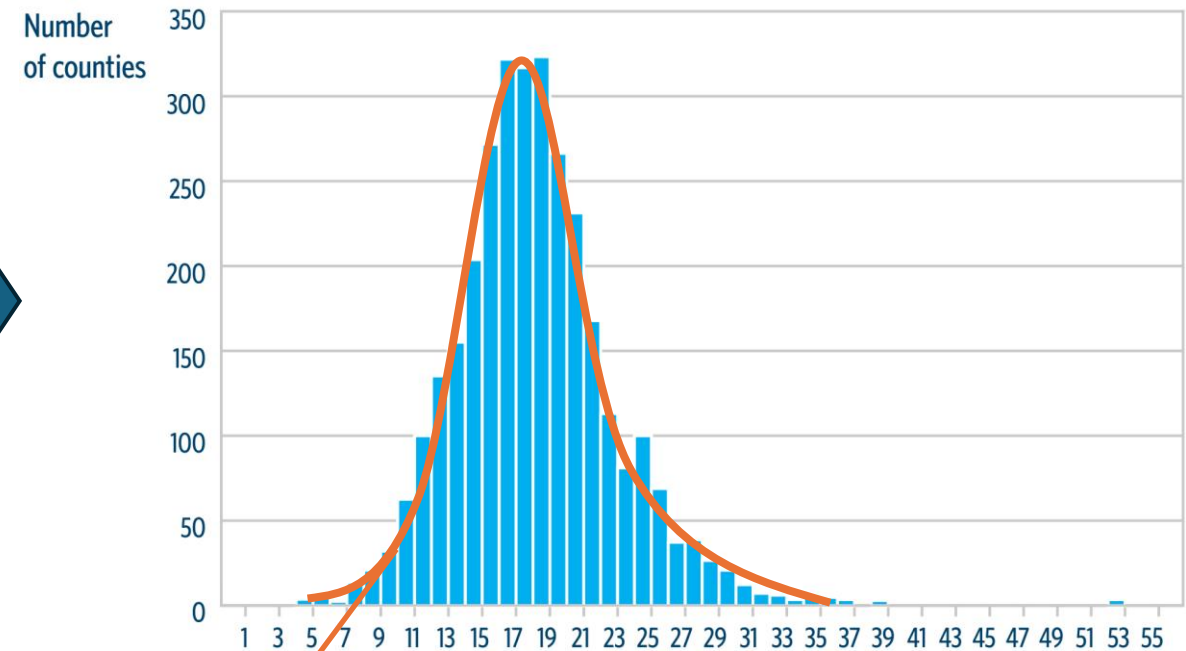
Item 1: Percentage of Population Aged 65 and Older



Percentage of seniors (aged 65 years and over) in each county

factfinder.census.gov/bkmk/table/1.0/en/PEP/2014/PEPAGESEX/0100000US.05000.003?slice=GEO-0500000US01005

Item 1: Percentage of Population Aged 65 and Older



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How to represent this shape with a math function ?

Normal Distribution (Gaussian Distribution)

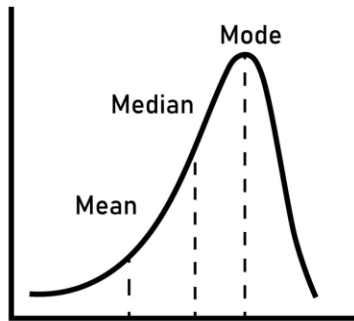
- **Definition:** A continuous probability distribution that is symmetrical around its mean, with most observations clustering around the central peak and the probabilities for values further away from the mean tapering off equally in both directions. This creates the classic "bell curve." In a perfect normal distribution, the mean, median, and mode are all equal.

- Formula (Probability Density Function):

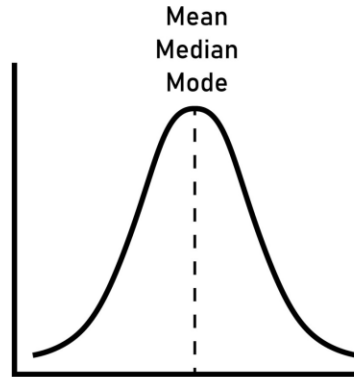
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Skewness

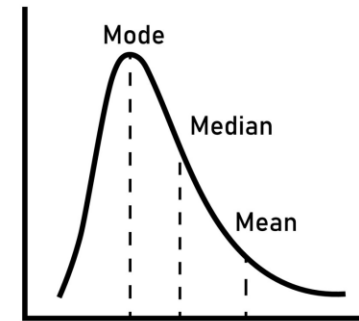
Negatively Skewed



Symmetric



Positively Skewed



Formula (Sample Skewness / Fisher-Pearson Coefficient):

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Independence vs. Dependence in Data

- **Independence**

- **Definition:** Two sets of data (or two variables) are independent if the value of one data point tells you absolutely nothing about the value of another. They do not affect each other.
- **Example:** If you measure the heights of randomly selected people in New York and compare them to the heights of randomly selected people in London, the measurements are independent

Independence vs. Dependence in Data

- **Dependence**

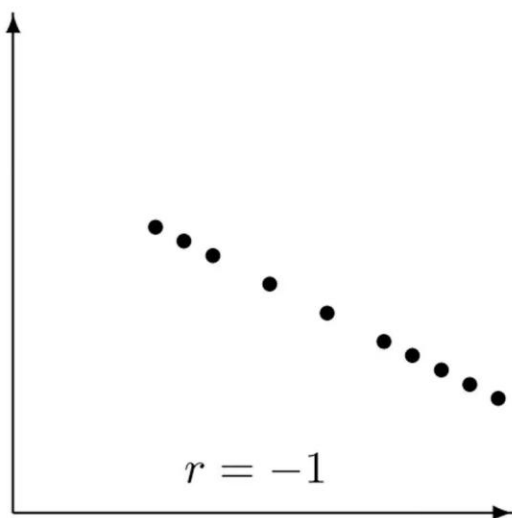
- **Definition:** Data is dependent if the value of one observation is explicitly linked to, or influenced by, another observation.
- **Example:** If you measure a group of patients' blood pressure **before** they take a new medication, and then measure those same patients' blood pressure **after** taking it. The "after" measurement is highly dependent on that individual's "before" baseline.

Correlation

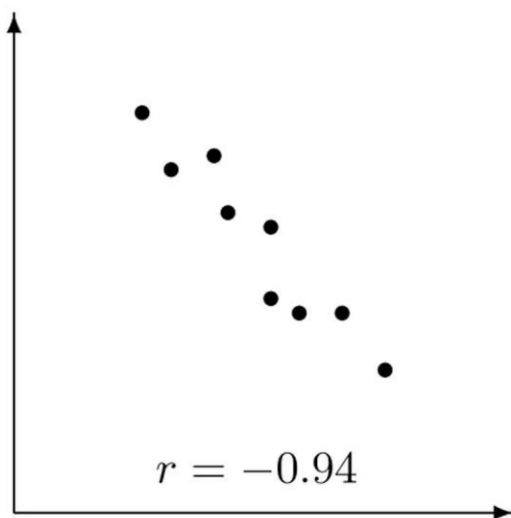
- **Correlation** is a specific statistical measurement of *how* two variables are related to each other. If two variables are correlated, they are mathematically dependent.
- **Definition:** Correlation measures the strength and direction of a linear relationship between two variables.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

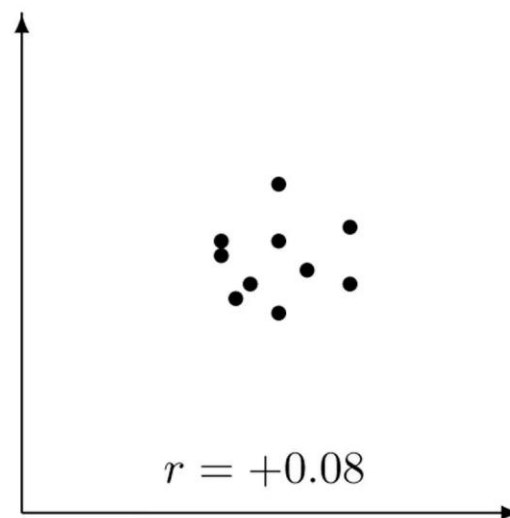
Correlation



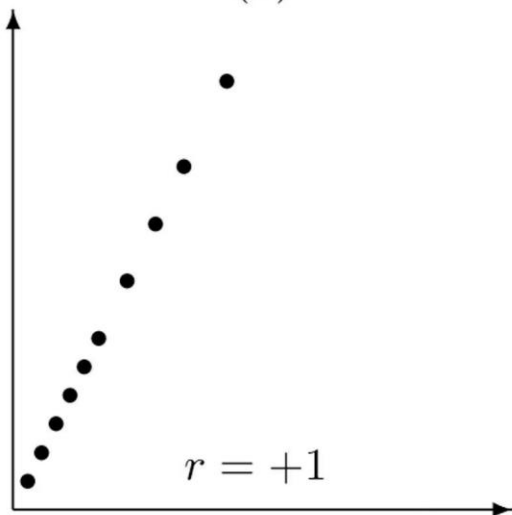
(a)



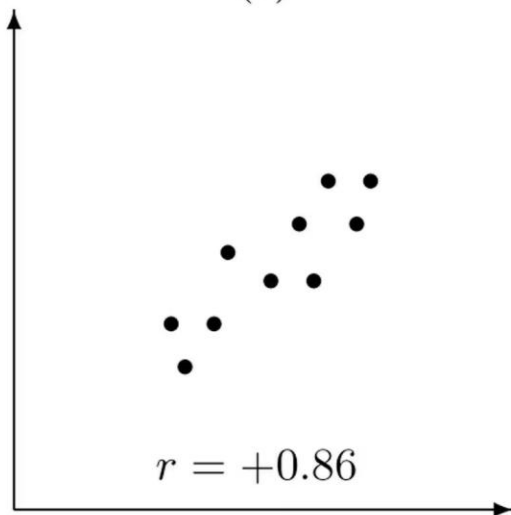
(b)



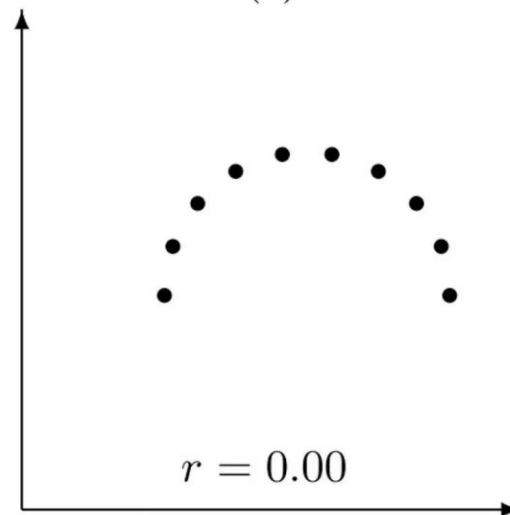
(c)



(d)



(e)



(f)

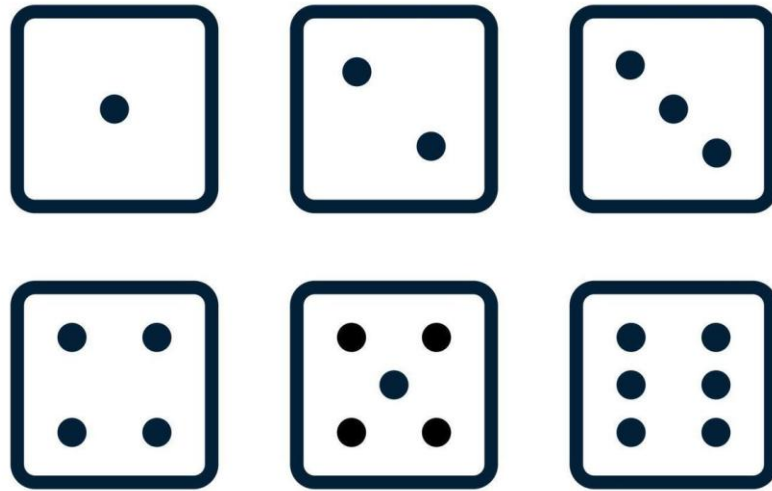
Minimum Probability

Probability Fundamentals & Events

- **Sample Space (Ω):** The set of all possible outcomes of an experiment.
- **Event (A):** A subset of the sample space (a specific outcome or collection of outcomes).
- **Probability ($P(A)$):** A number between 0 and 1 that represents the likelihood of an event occurring, where 0 means impossible and 1 means certain.

Example,

- **Sample Space (Ω):** Rolling a standard die. $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- **Event (A):** Rolling an even number. $A = \{2, 4, 6\}$.
- **Probability ($P(A)$):** 3 even outcomes out of 6 total outcomes $= \frac{3}{6} = 0.5$.



Conditional Probability

- **Definition:** The probability of an event (A) occurring, *given* that another event (B) has already occurred. It restricts the sample space to only the outcomes where B happens.
- **Formula:**

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

(Assuming $P(B) > 0$)

Example,

- **Scenario:** Drawing a single card from a standard 52-card deck.
- **Event A (Drawing a King):** There are 4 Kings. $P(A) = \frac{4}{52} = \frac{1}{13}$.
- **Event B (Drawing a Face Card):** There are 12 Face Cards (Jacks, Queens, Kings). $P(B) = \frac{12}{52}$.
- **Conditional Probability:** What is the probability you drew a King, *given* that you know you drew a Face Card?
 - Using the formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
 - The intersection ($A \cap B$) is just the 4 Kings.

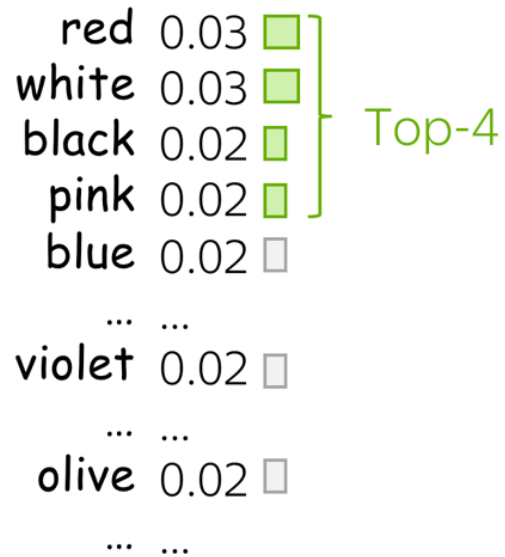
$$P(A|B) = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3}$$

Example,

Top-K for a flat distribution: not enough

The dress color was _____

$P(* \mid \text{The dress color was})$

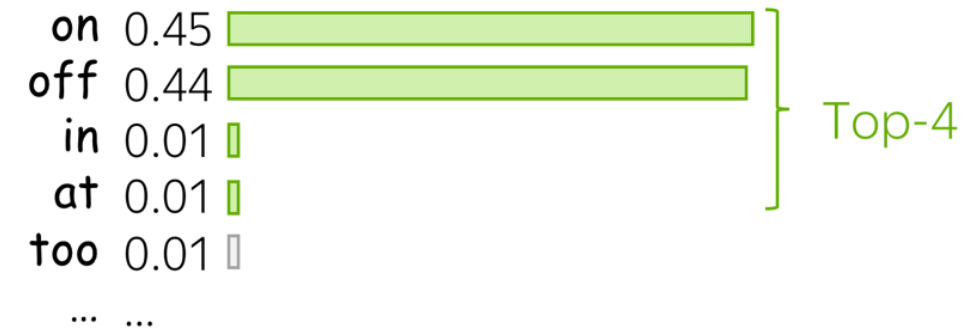


Top-K for a peaky distribution: too many

The light was _____

$P(* \mid \text{The light was})$

get probability
distribution



Independence

- **Definition:** Two events are independent if the occurrence of one does not change the probability of the other occurring.

- Mathematical Condition:

$$P(A \cap B) = P(A)P(B)$$

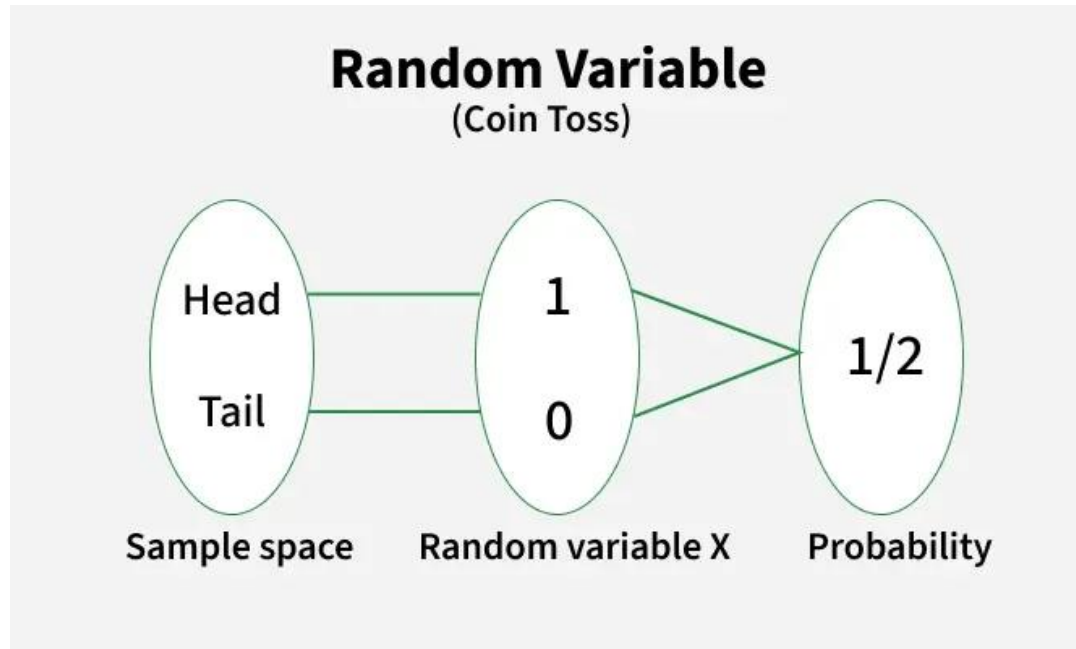
- In terms of Conditional Probability:

$$P(A|B) = P(A)$$

Random Variables: Discrete vs. Continuous

- A random variable (X) is a numerical description of the outcome of a statistical experiment.
 - **Discrete Random Variables:** Can take on a countable number of distinct values (e.g., rolling a die, counting the number of emails received). Described by a Probability Mass Function (PMF).
 - **Continuous Random Variables:** Can take on any value within an interval or continuum (e.g., exact height, time taken to run a mile). Described by a Probability Density Function (PDF).

Random Variable vs. Programming Variable



Stochastic

$$x = 0$$

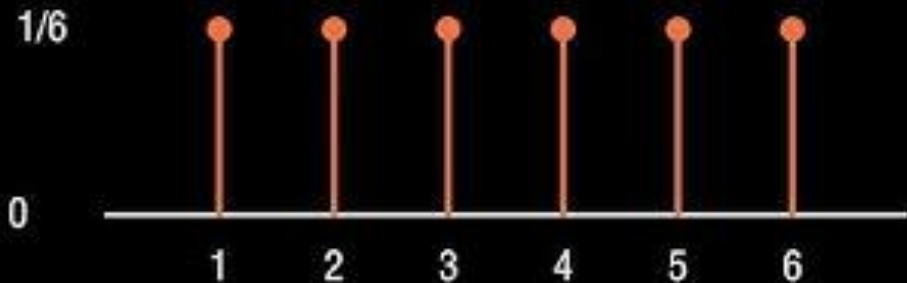
Deterministic

PMF, PDF, and CDF

Probability Distributions: Visual Primer

Discrete

Probability Mass Function

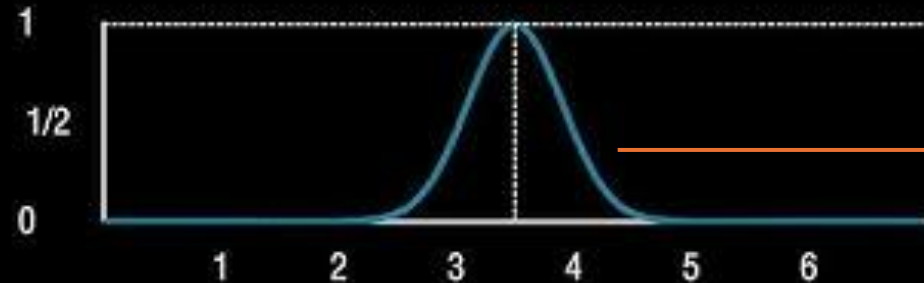


Cumulative Distribution Function

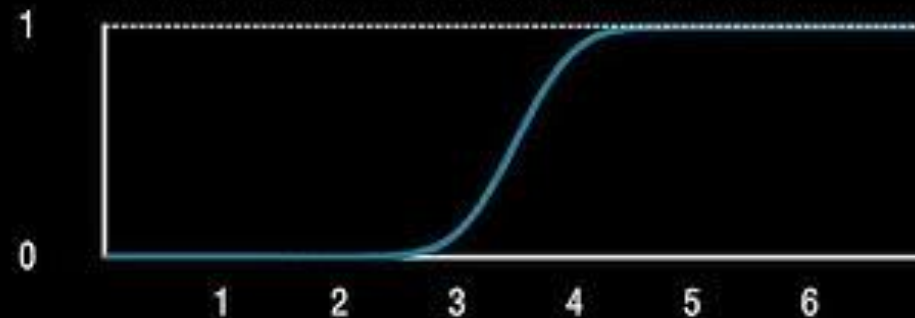


Continuous

Probability Density Function



Cumulative Distribution Function



Normal distribution is an example PDF.

Expected Value ($E[X]$ or μ)

- **Definition:** The long-run average value of repetitions of the experiment it represents; the "center of mass" of the distribution

- Discrete Formula:

$$E[X] = \sum_x xP(X = x)$$

- Continuous Formula:

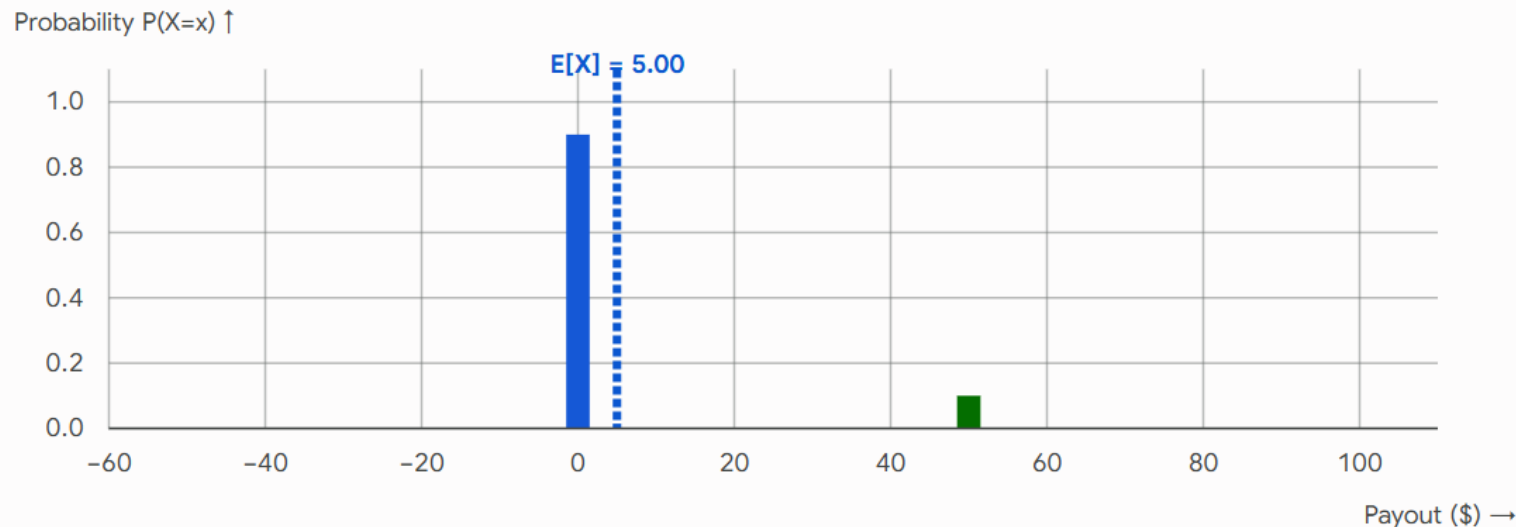
$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Example,

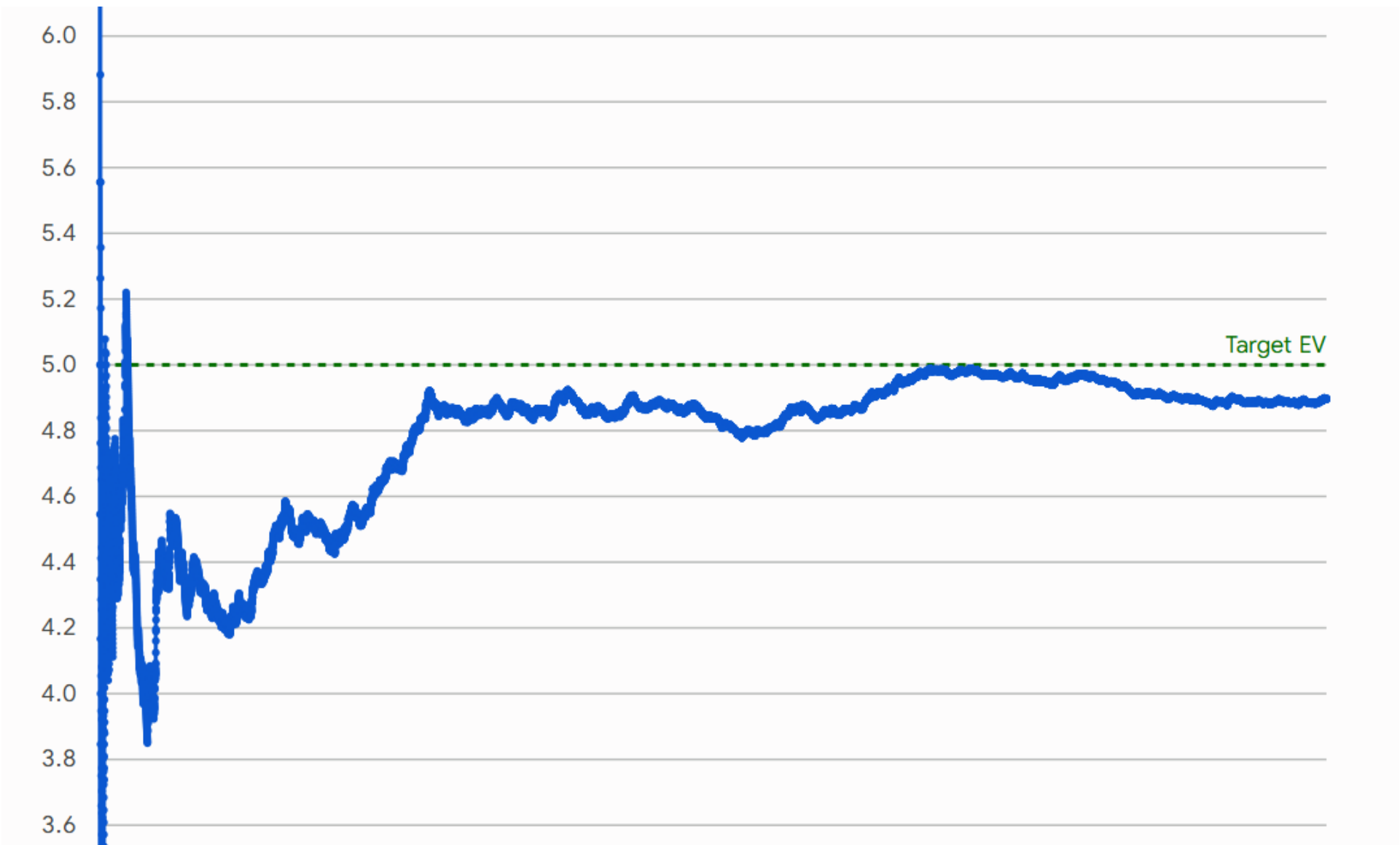
- Playing a carnival game where you have a 10% chance to win \$50 and a 90% chance to win \$0.
 - $E[X] = (0.10 * 50) + (0.90 * 0) = \5
 - On average, over many games, a ticket is "worth" \$5.

Probability Mass Function Visualizer

■ Loss Outcome ■ Win Outcome



Example,



Variance ($Var(X)$ or σ^2)

- **Definition:** The expected value of the squared deviation from the mean. It measures how spread out the numbers are.

- Formula:

$$Var(X) = E[(X - E[X])^2]$$

- Alternative (Computational) Formula:

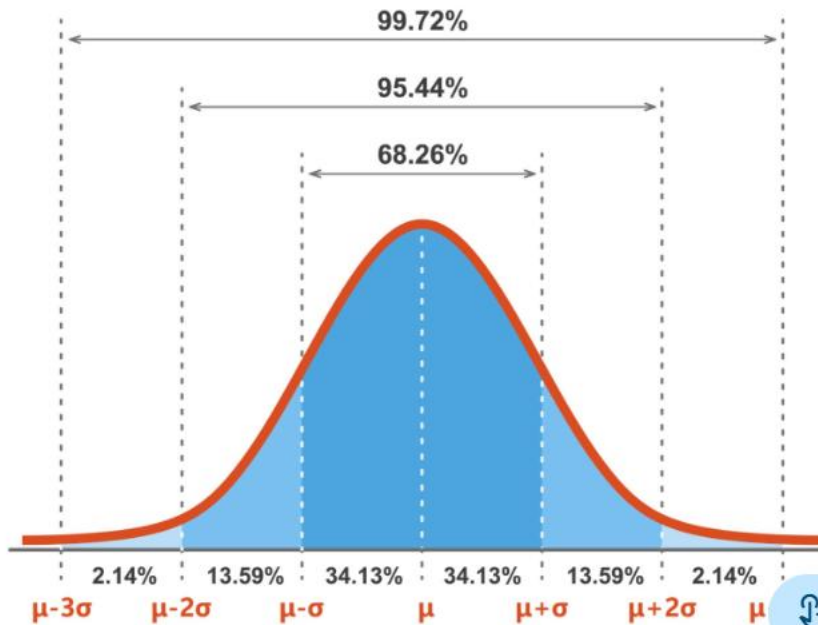
$$Var(X) = E[X^2] - (E[X])^2$$

Core Distributions

Gaussian (Normal) Distribution

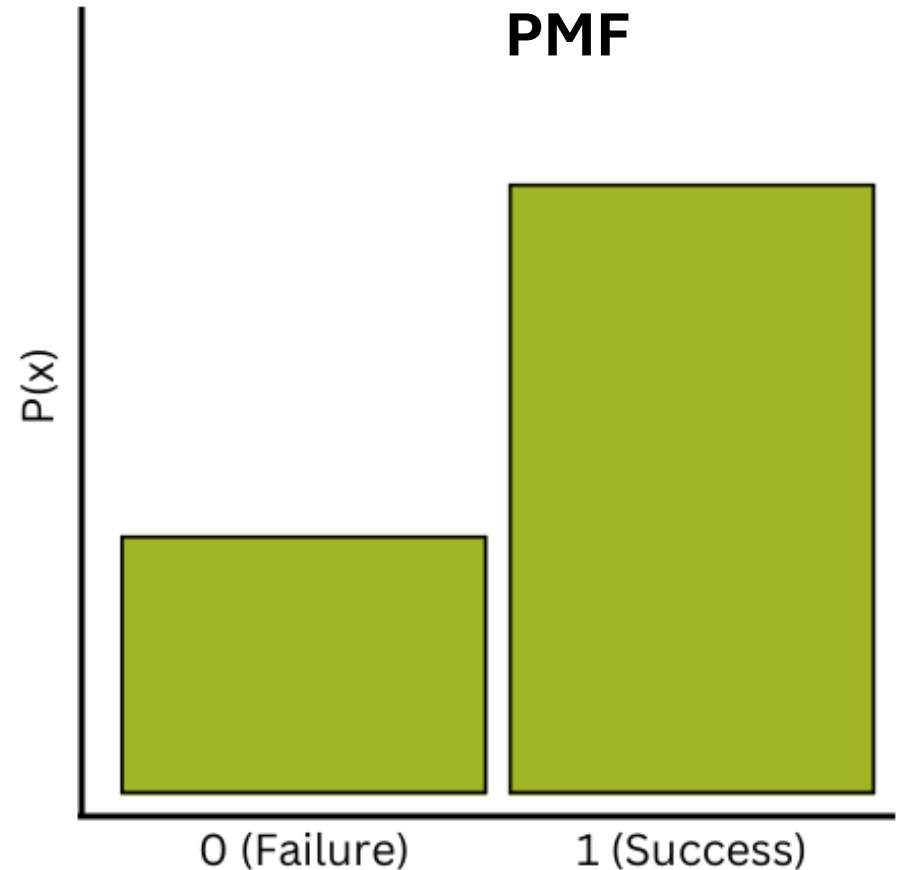
- Parameters: μ (mean), σ^2 (variance).
- PDF Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Bernoulli Distribution

- **Use Case:** A single trial with exactly two possible outcomes (Success/Failure, 1/0).
- **Parameter:** p (probability of success).
- **Expected Value:** p
- **Variance:** $p(1 - p)$



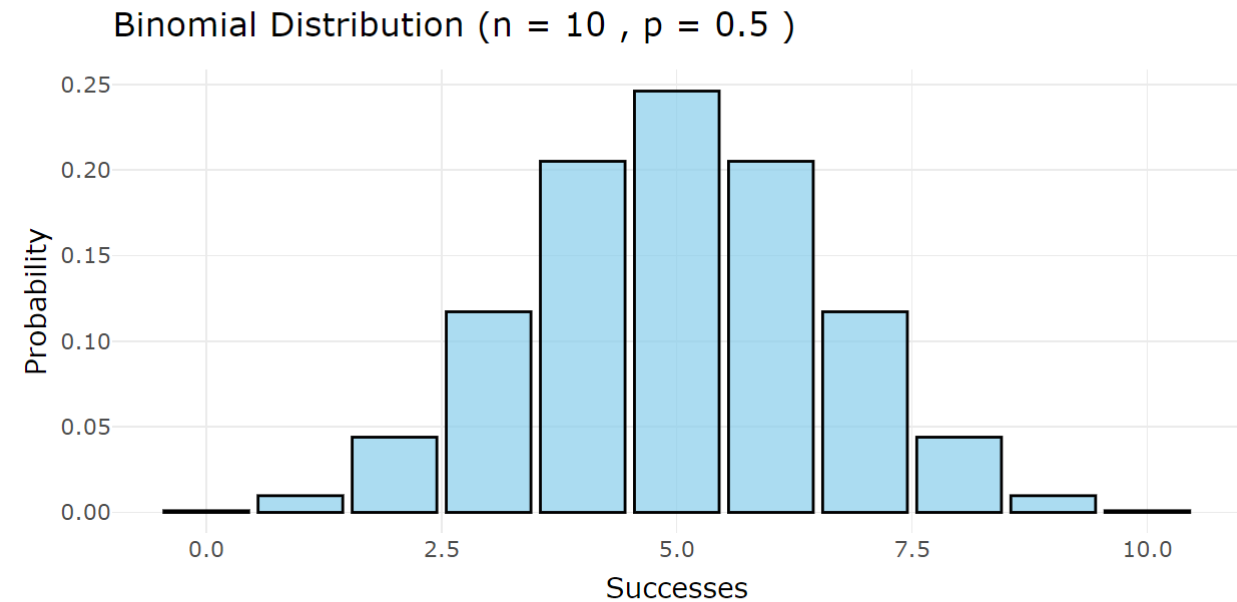
Binomial Distribution

- **Use Case:** The number of successes in n independent Bernoulli trials.
- **Parameters:** n (number of trials), p (probability of success per trial).
- **Expected Value:** np
- **Variance:** $np(1 - p)$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}$$

(where ! represents a factorial, e.g., $4! = 4 * 3 * 2 * 1$)



Mean & Variance

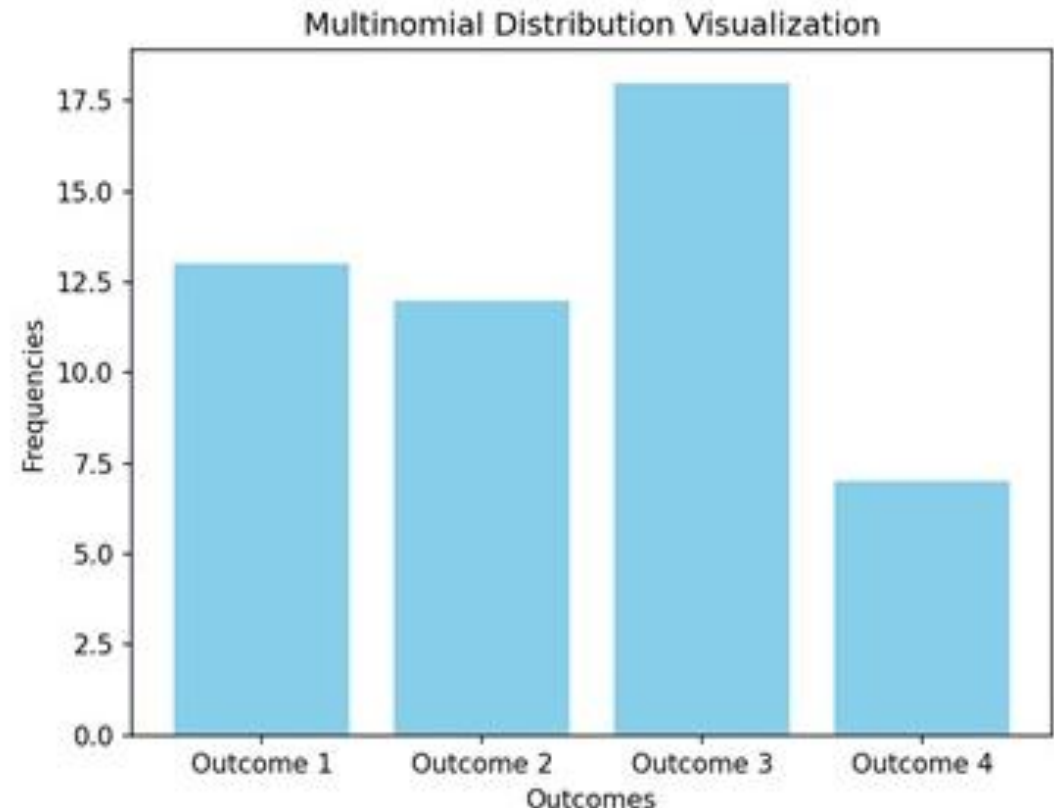
Mean: 5
Variance: 2.5
Std. Dev.: 1.58

Categorical Distribution (Multinoulli)

- **Use Case:** A single trial where the outcome is exactly one of k possible categories (e.g., rolling a single 6-sided die). It is a generalization of the Bernoulli distribution for more than two outcomes.
- **Parameters:** $p_1, p_2, \dots, p_k$ (where the sum of all probabilities equals 1)

$$P(\mathbf{x}) = \prod_{i=1}^K p_i^{x_i}$$

PMF



Bayes' Rule

Bayes' Rule

- **Definition:** A mathematical formula used to update the probabilities of hypotheses when given new evidence. It relates current (posterior) probability to prior probability.

- Formula:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- The Components:

- $P(A|B)$ (**Posterior**): The probability of hypothesis A being true, given the evidence B .
- $P(B|A)$ (**Likelihood**): The probability of observing the evidence B if hypothesis A is true.
- $P(A)$ (**Prior**): The initial probability of hypothesis A being true before considering the new evidence.
- $P(B)$ (**Marginal**): The total probability of observing the evidence B under all possible hypotheses.

Example,

- **Scenario:** You are building a spam filter.
 - $P(\text{Spam}) = 0.40$ (40% of all emails you get are spam). This is your **Prior**.
 - $P(\text{"Free"}|\text{Spam}) = 0.80$ (80% of spam emails contain the word "Free"). This is the **Likelihood**.
 - $P(\text{"Free"}|\text{Not Spam}) = 0.10$ (10% of regular emails also use the word "Free").
- **Question:** An email arrives containing the word "Free". What is the probability it is actually spam? $P(\text{Spam}|\text{"Free"})$?

Example,

- **Step 1: Calculate Marginal Probability $P(\text{"Free"})$ (The Denominator)**

You must find the total probability of seeing the word "Free" in *any* email:

$$P(\text{"Free"}) = (0.80 \times 0.40) + (0.10 \times 0.60) = 0.32 + 0.06 = 0.38$$

$$P(\text{Free}) = P(\text{Free}|\text{Spam}) * P(\text{Spam}) + P(\text{Free}|\text{Not Spam})$$

Example,

- Step 2: Apply Bayes' Formula

$$P(\text{Spam} | \text{"Free"}) = \frac{P(\text{"Free"} | \text{Spam}) \times P(\text{Spam})}{P(\text{"Free"})}$$

$$P(\text{Spam} | \text{"Free"}) = \frac{0.80 \times 0.40}{0.38} = \frac{0.32}{0.38} \approx 0.842$$

Even though regular emails sometimes use the word "Free", if you see that word, there is an 84.2% chance the email is spam.

Linear Algebra

The Building Blocks of Data

Scalars

- **Definition:** A single number that represents magnitude only. It has a 0-dimensional shape.
- **Example:** A learning rate (0.01), a loss value, or a specific metric output like a single reconstructed signal at a given timestamp.
- **Notation:** Usually a lowercase italic letter (x , a).

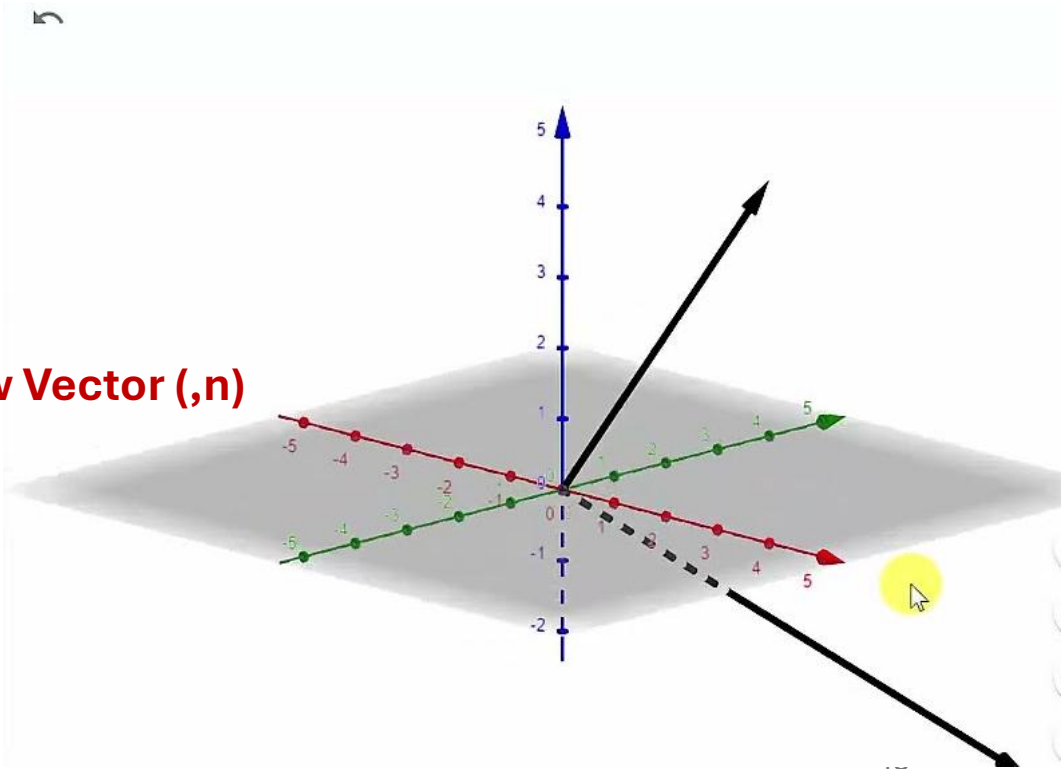
Vectors

- **Definition:** A 1D array of numbers. It represents both magnitude and direction in space.
- **Example:** A row in a dataset, or a coordinate x, y, z .
- **Notation:** A lowercase bold letter ($\boldsymbol{x}$).

48px [

		Column Vector (m,)		
User Name	User ID	User Phone	Status	
Ann Doe	MG124562	628-9052	Approved	
Brad Stant	CW719413	948-6074	Denied	
John Herwit	BC512441	767-0793	Approved	
Lydia Work	SP900124	211-5309	Denied	
Allison Dias	ZY857353	215-5309	Approved	

Row Vector (,n)



Matrices

- **Definition:** A 2D grid of numbers, organized by rows and columns.
- **Example:** A grayscale image (height * width), or a batch of inputs passing through a neural network.
- **Notation:** An uppercase bold letter (A , W)



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	227	229	228	227	87	71	201	
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	168	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

What the computer sees

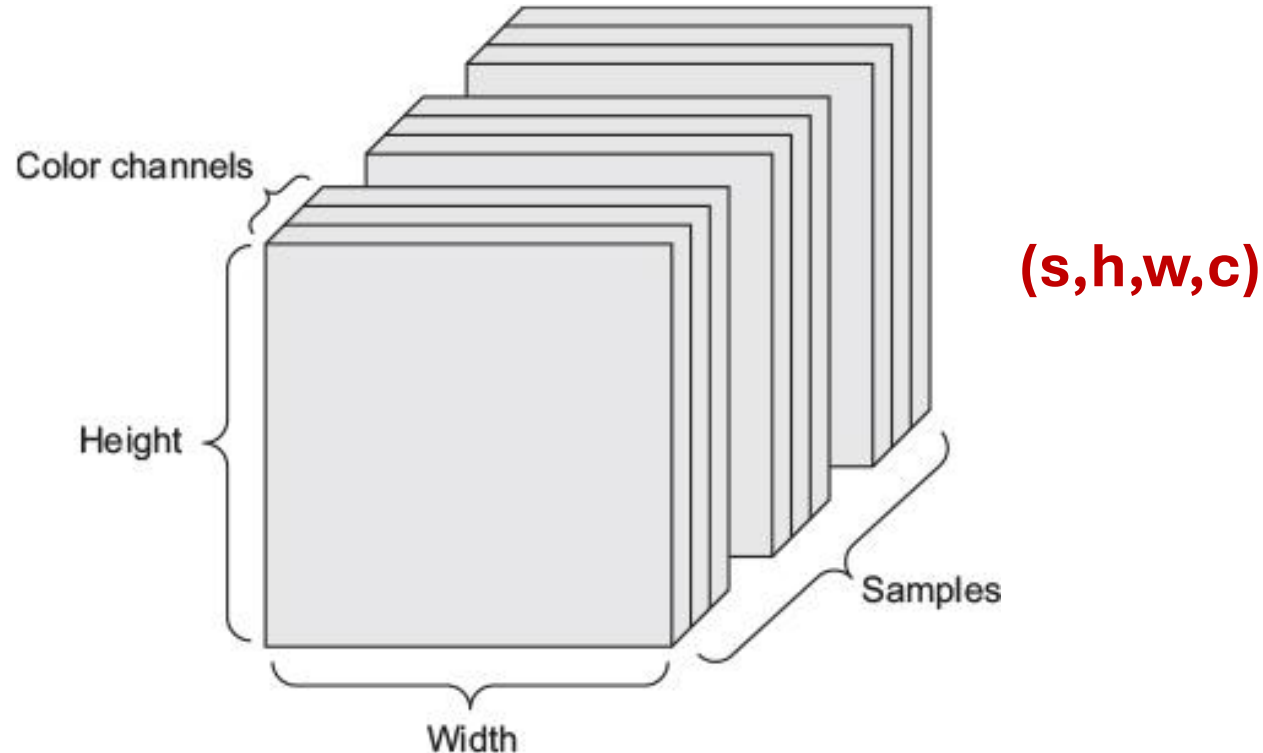
157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	227	229	228	227	87	71	201	
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	168	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

(m,n)

An image is just a matrix of numbers [0,255]!
i.e., 1080x1080x3 for an RGB image

Tensors

- **Definition:** An n -dimensional array of numbers. Scalars (0-D), vectors (1-D), and matrices (2-D) are technically all sub-types of tensors.
- **Example:** A batch of color images is a 4D tensor:
 $[batch_size, height, width, color_channels]$.



Together

Scalar Vector Matrix Tensor

1

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$\begin{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} & \begin{bmatrix} 3 & 2 \end{bmatrix} \\ \begin{bmatrix} 1 & 7 \end{bmatrix} & \begin{bmatrix} 5 & 4 \end{bmatrix} \end{bmatrix}$

(11)

$\begin{bmatrix} 5 & 3 & 7 \end{bmatrix}$

$\begin{bmatrix} 5 \\ 1.5 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 4 & 19 & 8 \\ 16 & 3 & 5 \end{bmatrix}$

$\begin{matrix} & A & B & C \\ 1 & a & b & c \\ 2 & d & e & f \\ 3 & g & h & i \\ 4 & j & k & l \\ 5 & m & n & o \\ 6 & p & q & r \\ 7 & s & t & u \\ 8 & v & w & x \\ 9 & y & z & \end{matrix}$

Scalar

Row Vector
(shape 1×3)

Column Vector
(shape 3×1)

Matrix

Tensor

Matrix Operations

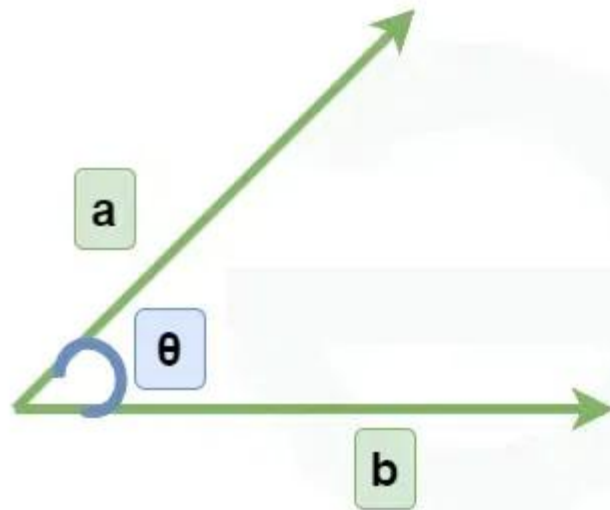
Addition

- **Definition:** Adding two matrices together element by element. They *must* be the exact same shape.
- **Example:** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$

Dot Product

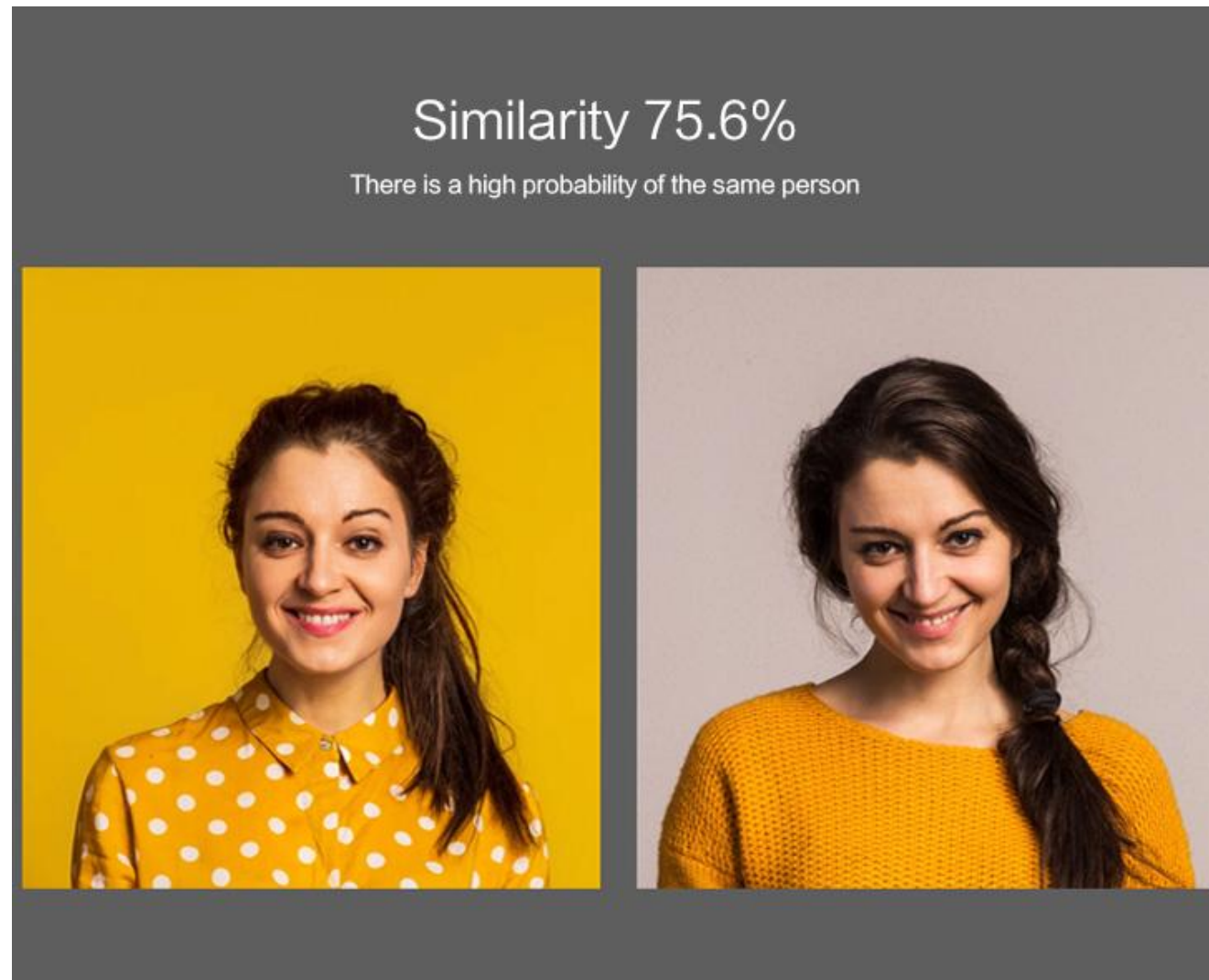
- **Definition:** An algebraic operation that takes two equal-length sequences of numbers (vectors) and returns a single scalar number. It measures how much one vector "points" in the same direction as another.
- **Formula:** $\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$

Dot Product



Dot Product of Vectors **a** and **b**
$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos(\theta)$$

Dot Product

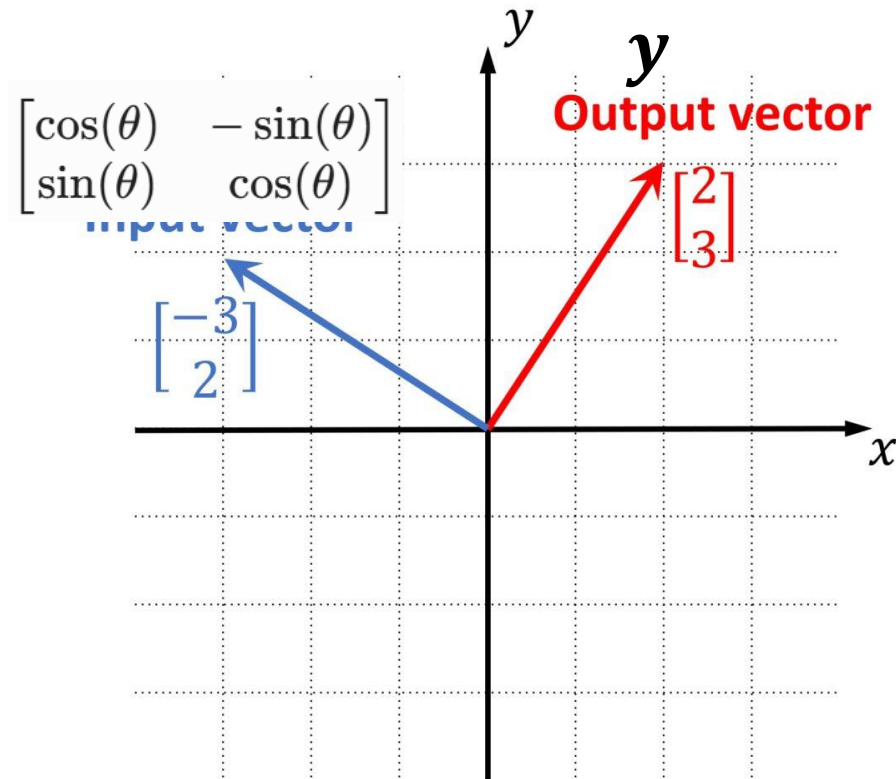


Vector-matrix multiplication

To multiply a matrix by a vector, you calculate the **dot product** of each **row** of the matrix with the entire column vector.

$(m,)$ (m,n) $(n,)$

$$\mathbf{y} = \mathbf{W}\mathbf{x}$$



$\mathbf{W}$ is rational matrix

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Example,

$$\mathbf{W} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

The Calculation:

- **Top output (y_1):** Take the dot product of the **top row** of $\mathbf{W}$ and the vector $\mathbf{x}$.

$$y_1 = (1 \times 5) + (2 \times 6) = 5 + 12 = 17$$

- **Bottom output (y_2):** Take the dot product of the **bottom row** of $\mathbf{W}$ and the vector $\mathbf{x}$.

$$y_2 = (3 \times 5) + (4 \times 6) = 15 + 24 = 39$$

The Result:

$$\mathbf{y} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

2D Rotation

The standard formula for a 2D rotation matrix by an angle θ is:

$$\mathbf{R} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

2D Scaling

This matrix multiplies the x and y coordinates by specific factors (s_x and s_y).

- **Matrix:**

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Matrix Multiplication

$$y = ABx$$



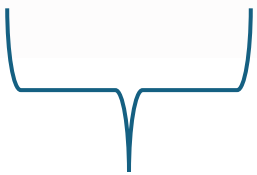
C

The composition of linear transformations.

- **Definition:** The composition of linear transformations. To multiply Matrix **A** by Matrix **B**, you take the **dot product** of the rows of **A** with the columns of **B**.
- **Rule:** The inner dimensions must match. You can multiply a $(m \times n)$ matrix by an $(n \times p)$ matrix, resulting in an $(m \times p)$ matrix.

Linear Equations

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

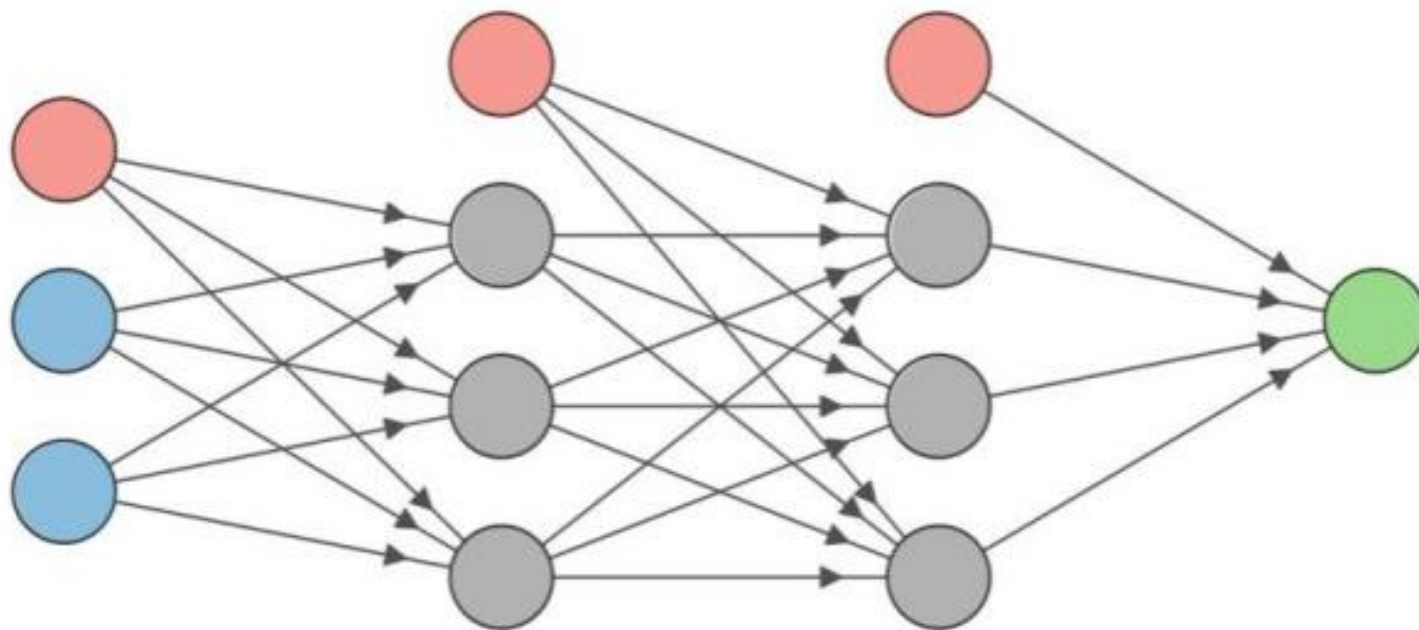


Vector-Matrix
Multiplication

A
Li

Deep neural network (nonlinear regression)

$$F(x) = w_3^T \sigma(W_2^T \sigma(W_1^T x + b_1) + b_2) + b_3$$



Input Layer $\in \mathbb{R}^3$

Hidden Layer $\in \mathbb{R}^4$

Hidden Layer $\in \mathbb{R}^4$

Output Layer $\in \mathbb{R}^1$

Norms and Distances

Norms quantify the "size" or "length" of a vector, which is critical for calculating loss (how far off a prediction is from reality).

- L^1 Norm (Manhattan Distance): The sum of the absolute values of the vector. $\sum |x_i|$
- L^2 Norm (Euclidean Distance): The standard "straight-line" distance. The square root of the sum of the squared vector values. $\sqrt{\sum x_i^2}$

Example,

$$a = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\text{distance}(a, b) = \|a - b\| = \sqrt{2}$$

$$\|a - b\| = \sqrt{(4 - 3)^2 + (2 - 1)^2} = \sqrt{2}$$

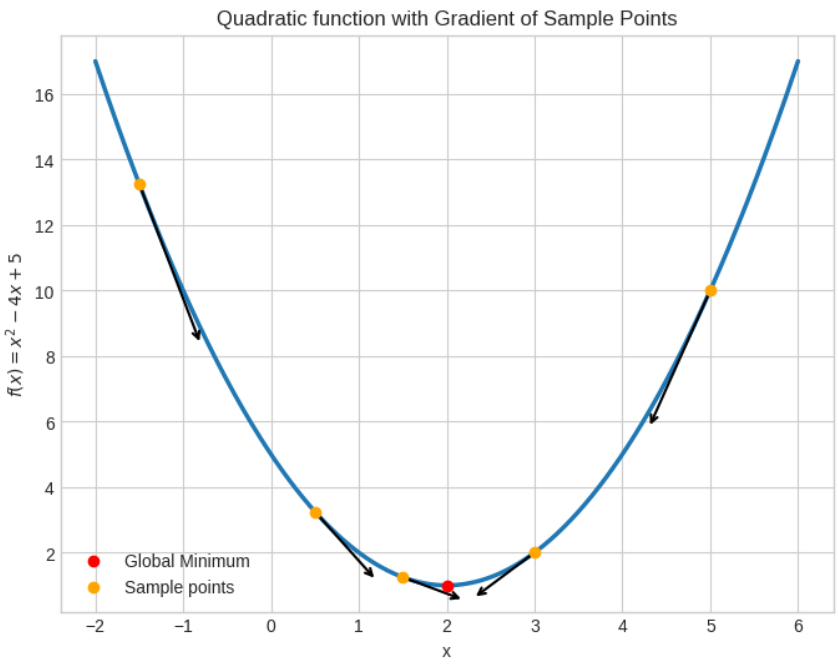
Vector Calculus

The engine of machine learning

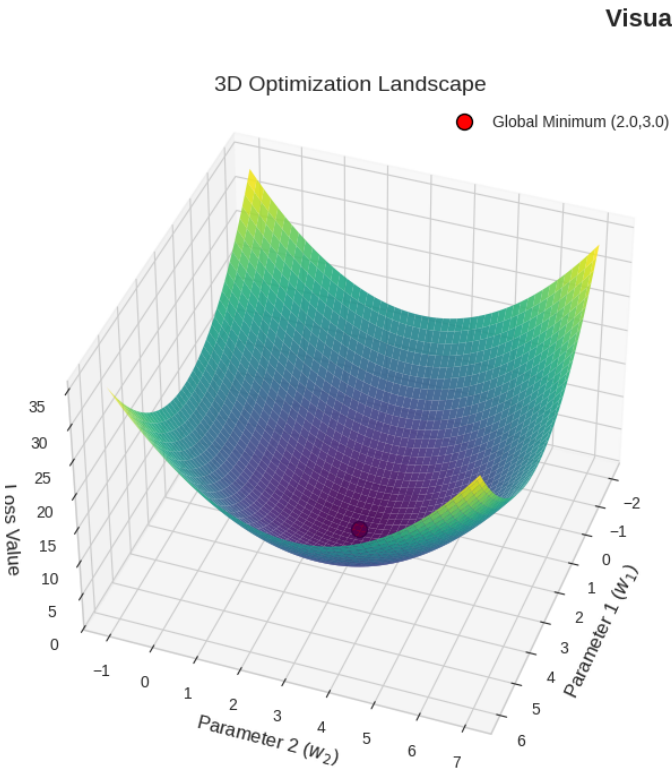
Functions and Graphs

- **Intuition:** A function maps ($y = f(x)$) inputs to outputs. In machine learning, the most important function is the **Loss Function** (or Cost Function), which measures how wrong a model's predictions are.
- **The Graph:** Imagine plotting every possible weight combination of a model on the x -axis and the resulting error on the y -axis. This creates a physical landscape of hills (high error) and valleys (low error). The entire goal of learning is to find the absolute lowest point in that valley.

Functions and Graphs

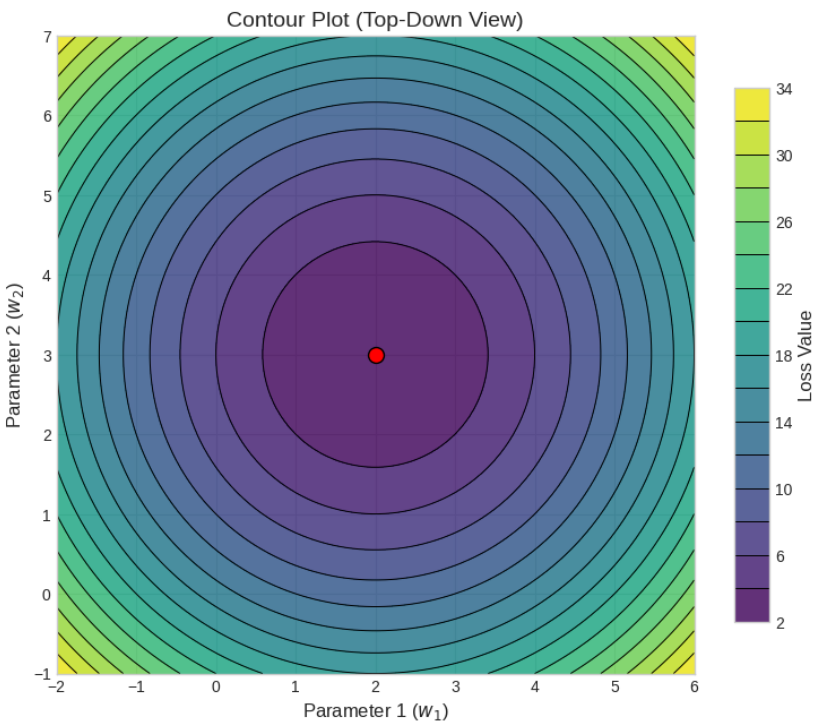


$$y = f(x_1)$$



$$y = f(x_1, x_2)$$

Visualizing a Loss Landscape



$$y = f(x_1, x_2, x_3)$$

How about an LLM with 1B parameters ?

Derivatives: The Slope ($\frac{df}{dx}$)

- **Definition:** The derivative measures the instantaneous rate of change of a function. It tells you the exact slope of the tangent line at any specific point on a curve.

- **Notation:**

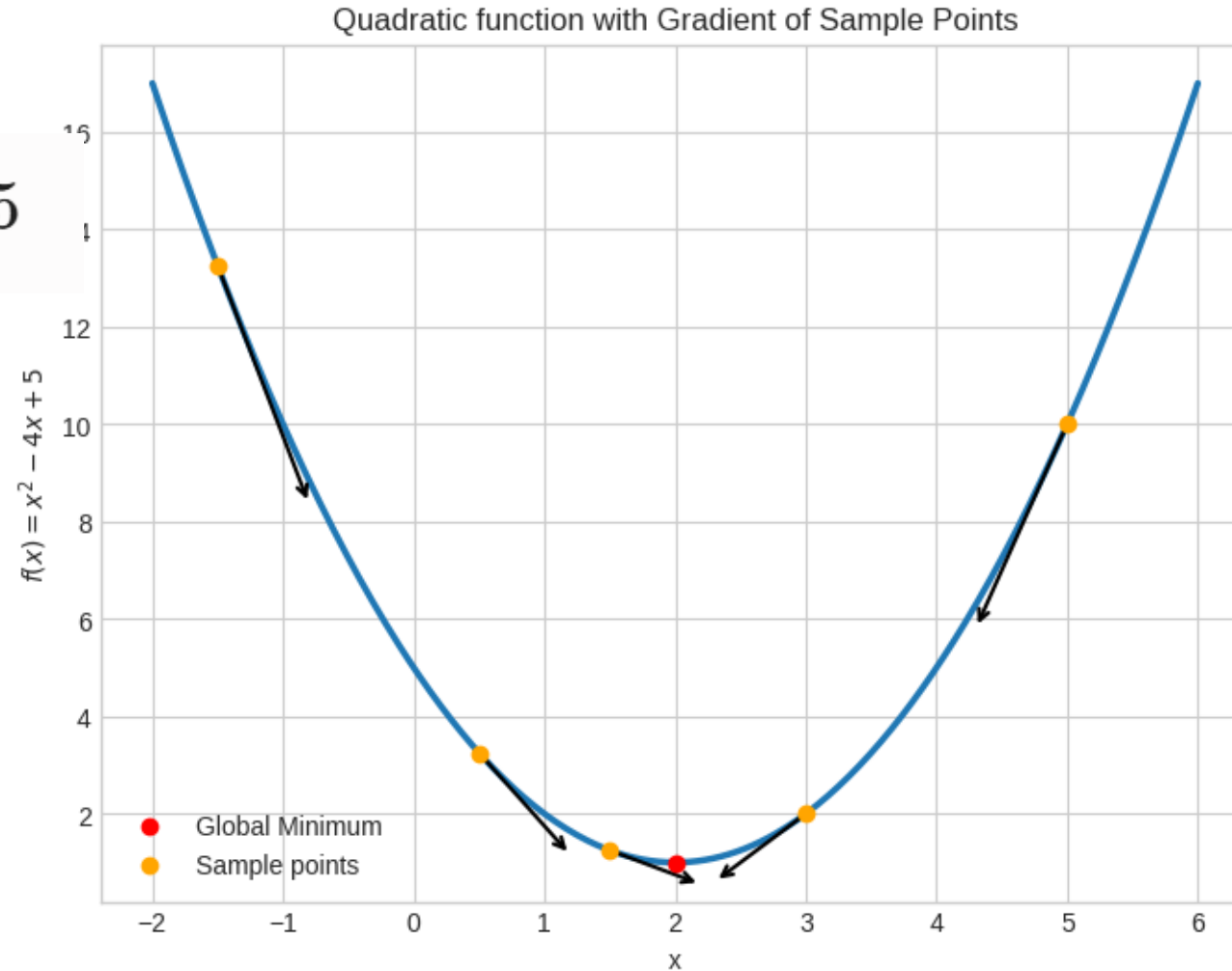
$$f'(x) = \frac{df}{dx}$$

- **ML Intuition:** If you are standing on the side of the "loss hill," the derivative tells you exactly how steep the hill is beneath your feet, and whether taking a step forward will move you up the hill (positive slope) or down the hill (negative slope).

Derivatives: The Slope ($\frac{df}{dx}$)

$$f(x) = x^2 - 4x + 5$$

$$f'(x) = 2x - 4$$



Partial Derivatives & Gradients

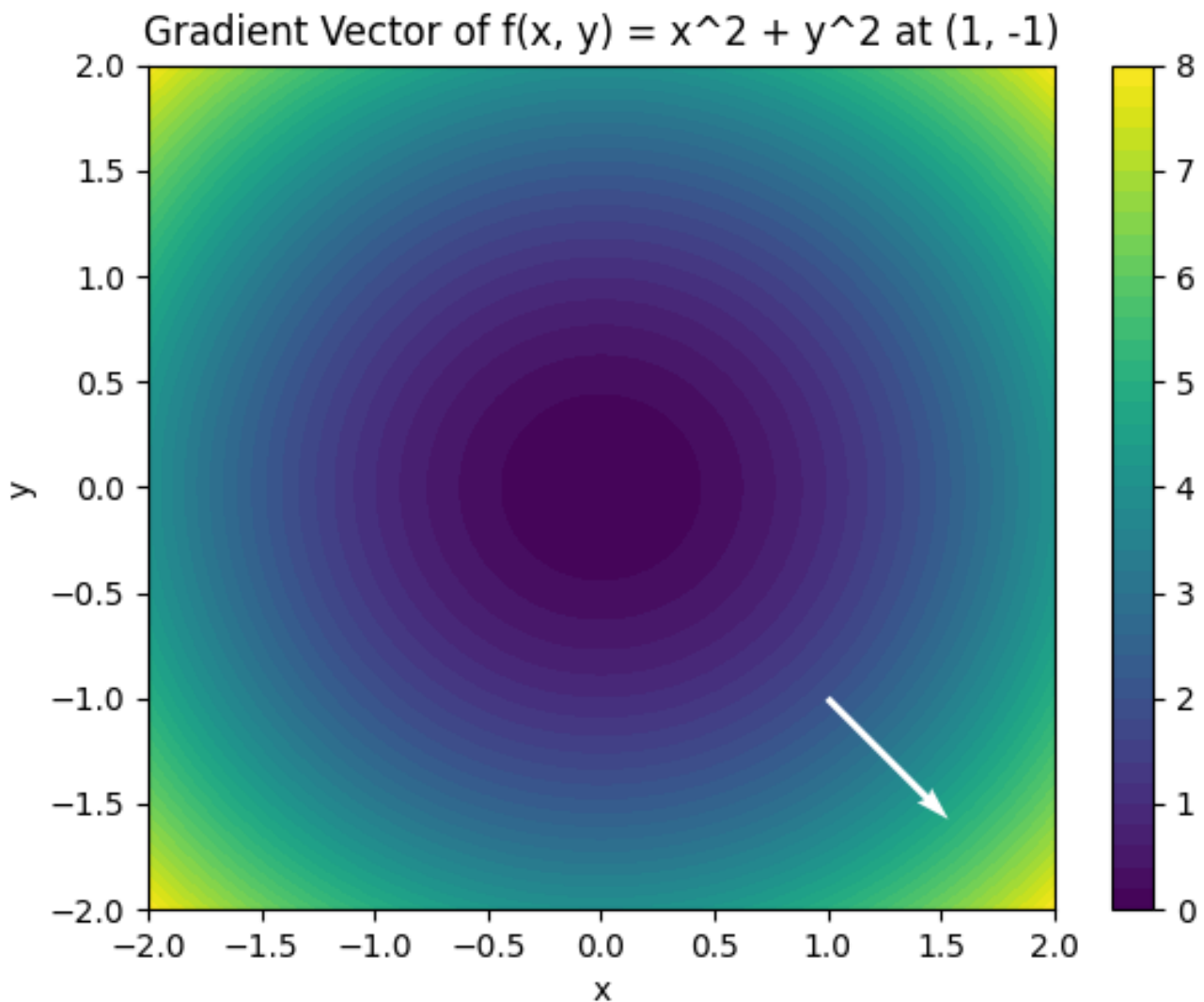
- **Partial Derivatives:** Real models don't have one weight; they have millions. A partial derivative ($\frac{\partial f}{\partial x_i}$) measures how the loss changes if you tweak *just one single weight*, pretending all the other millions of weights are frozen in place.
- **The Gradient (∇f):** * **Definition:** A vector that packages all the individual partial derivatives together.
 - **Crucial Property:** The gradient vector always points in the direction of **steepest ascent** (straight up the hill).

Gradients

$$f(x, y) = x^2 + y^2$$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\nabla f(1, -1) = \begin{bmatrix} 2(1) \\ 2(-1) \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$



The Chain Rule: The Engine of Backpropagation

$$y = (3x + 2)^2 \quad \longrightarrow \quad \frac{dy}{dx}?$$

$$u = 3x + 2$$

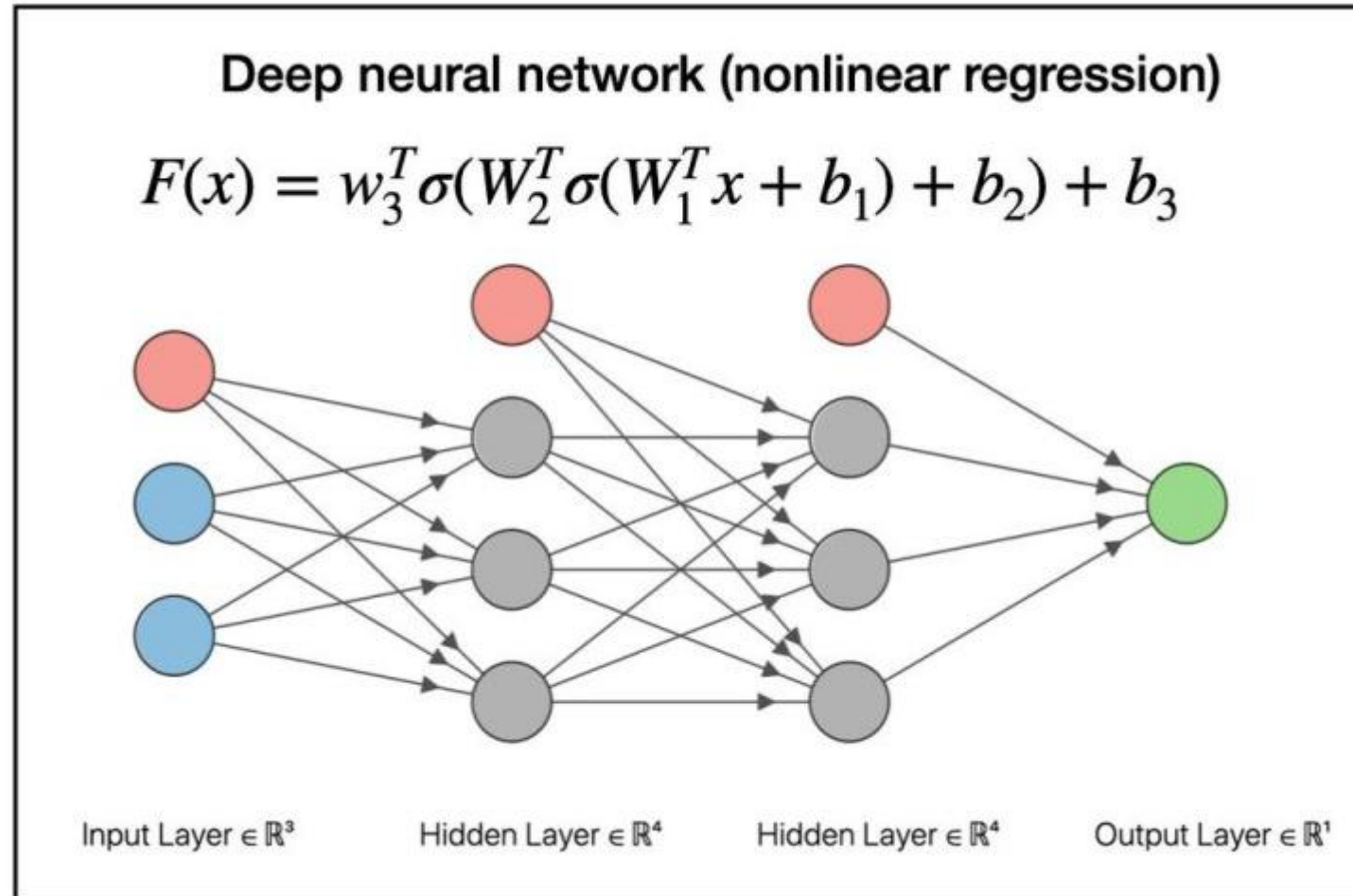
$$y = u^2$$

$$\frac{dy}{du} = 2u$$

$$\frac{dy}{dx} = \boxed{\frac{dy}{du}} \cdot \boxed{\frac{du}{dx}} = 6u = 6(3x + 2)$$

$$\frac{du}{dx} = 3$$

A Neural Network as a Complex Nested Function



Automatic Differentiation (also known as autograd)

Pytorch, TensorFlow, Jax can handle the gradient calculations.

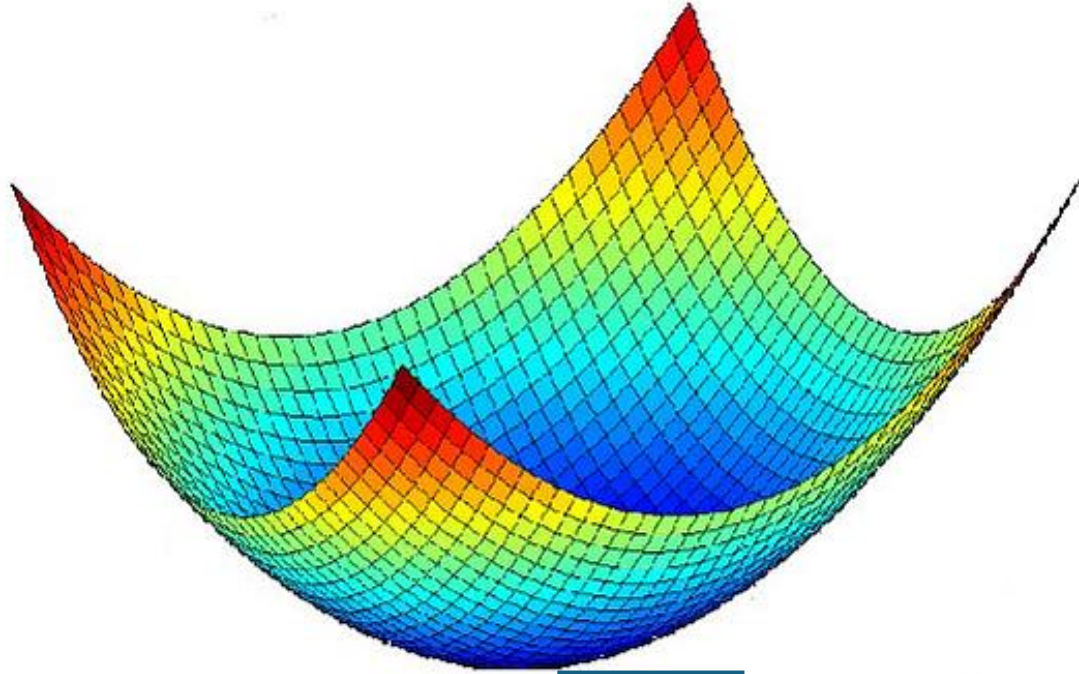
Just call ***backward()***



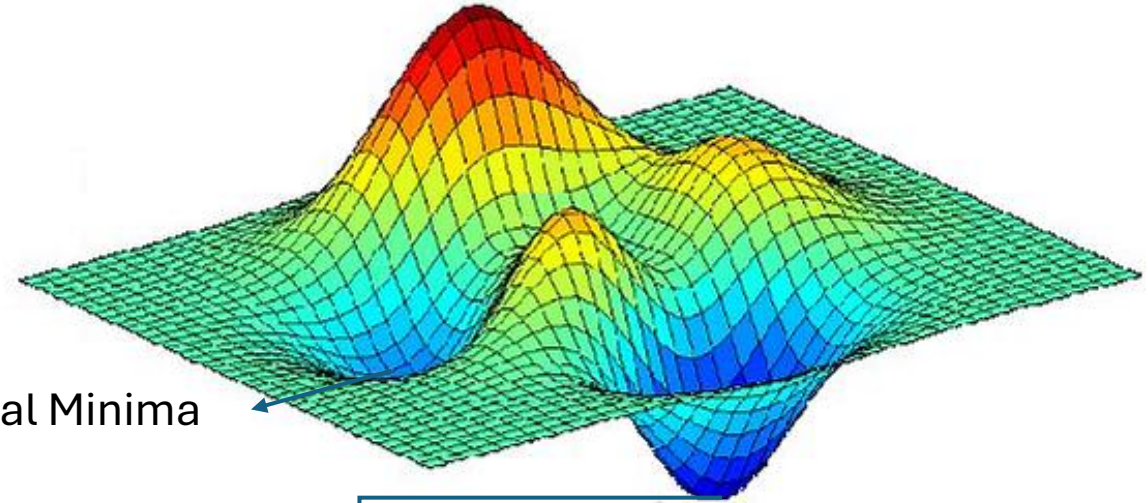
Convex vs. Non-Convex

- **Convex:** A perfect bowl shape. It has only one valley (the global minimum). If a problem is convex, gradient descent is mathematically guaranteed to find the absolute best solution
- **Non-Convex:** A rugged mountain range with many dips and peaks.
 - **Local Minima:** Shallow valleys that look like the bottom, but aren't the true bottom.
 - **Saddle Points:** Flat areas where the slope is zero, but it's not a valley at all.
 - **Deep Learning Reality:** The loss landscapes of deep neural networks are incredibly non-convex and chaotic. A massive area of AI research is designing optimizers (like Adam) that carry enough momentum to roll out of local minima and keep searching for the true global minimum.

Convex vs. Non-Convex



convex function



Local Minima

Global Minima

non-convex function

Any question/suggestion?